



## A+ LAB SERIES V5

### Lab 20: Writing Basic Scripts

Document Version: **2026-01-02**

| Material in this Lab Aligns to the Following |                                       |
|----------------------------------------------|---------------------------------------|
| CompTIA A+ (220-1201) Exam Objectives        |                                       |
| CompTIA A+ (220-1202) Exam Objectives        | 4.8: Explain the basics of scripting. |

## Contents

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## Introduction

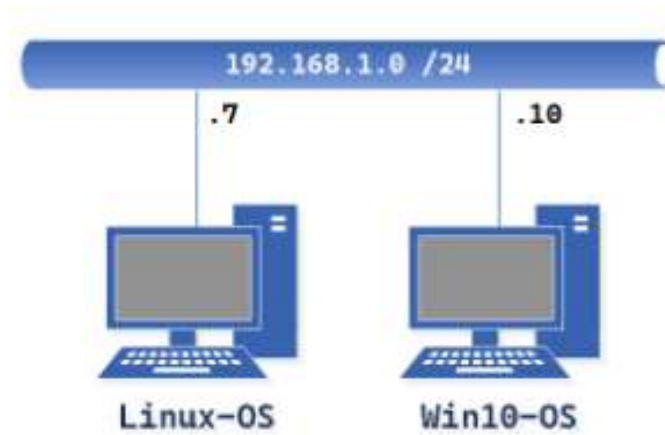
Scripts are primarily used to automate administration functions. This saves time and gives administrators a way to control how tasks are done, reducing the risk of errors and making tasks more time-efficient. In this lab, you will write scripts in different scripting languages to perform some simple tasks.

## Objective

In this lab, you will write basic scripts in different scripting languages to understand how each language handles automating tasks.

- Create a Windows Batch File (Script) using Notepad.
- Test the Windows Batch File using Command Prompt.
- Create a PowerShell script and test the PowerShell script.
- Create a BASH script using Linux Mint and test the BASH script.

## Lab Topology



## Lab Settings

The information in the table below will be needed to complete the lab. The task sections below provide details on the use of this information.

| Virtual Machine | IP Address        | Account<br>(if needed) | Password<br>(if needed) |
|-----------------|-------------------|------------------------|-------------------------|
| Linux-OS        | 192.168.1.7 / 24  | sysadmin               | NDGLabpass123!          |
| Win10-OS        | 192.168.1.10 / 24 | sysadmin               | NDGLabpass123!          |

## 1 Creating and Testing a Windows Batch File (Script)

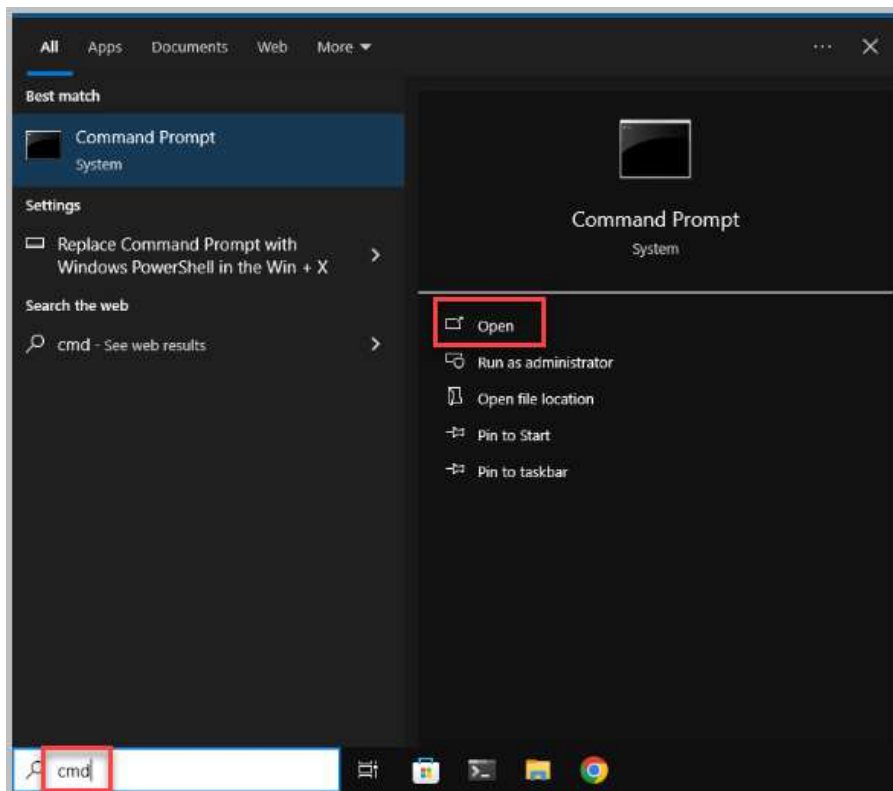
A batch file is a special kind of text file with a .bat extension that contains one or more commands a Windows command prompt understands and executes in sequence to perform tasks. Entering commands manually, one at a time, will achieve the same results, but a batch file simplifies retyping commands, saving time and reducing the risk of typos. Batch files are written in text editors. In this exercise, you will use Notepad to create the script file.

1. Open a console connection to the **Win10-OS** system.



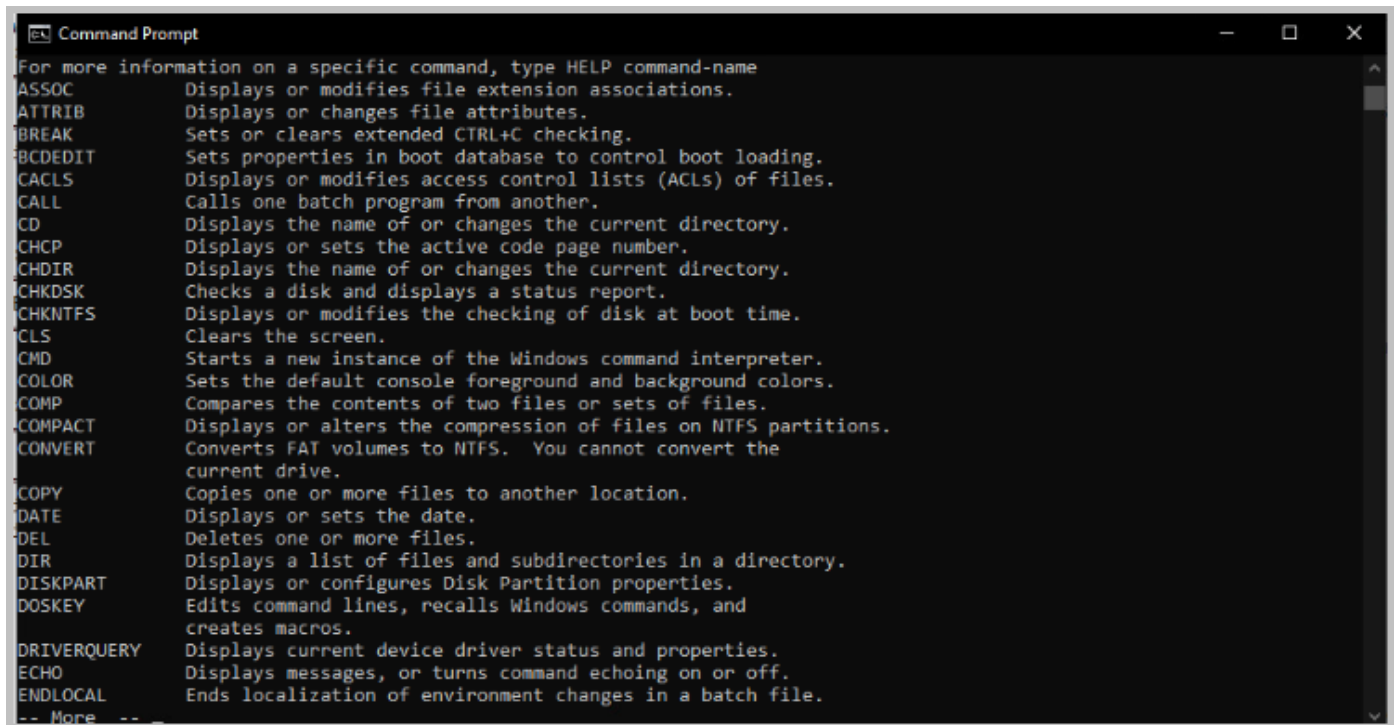
To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

2. In the **search** textbox located in the taskbar, type **cmd** in the search field. Select **Open**



3. At the prompt, type the command below, followed by pressing the **Enter** key. This output is a list of commands available in the Windows command-line interpreter. Review the list of commands identified below to understand the commands you will be exploring in the batch file. The **HELP | MORE** command shows all available commands and what each one does. In the list, review *echo*, *findstr*, *rem*, *systeminfo*, and *wmic*.

HELP | MORE

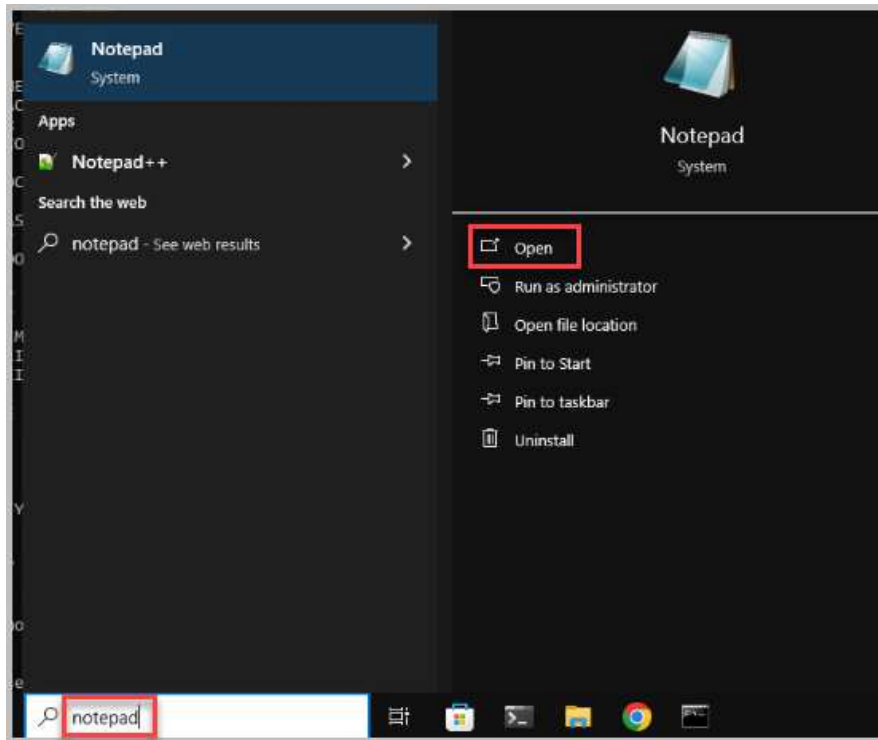


```
Command Prompt
For more information on a specific command, type HELP command-name
ASSOC      Displays or modifies file extension associations.
ATTRIB     Displays or changes file attributes.
BREAK      Sets or clears extended CTRL+C checking.
BCDEDIT    Sets properties in boot database to control boot loading.
CACLS      Displays or modifies access control lists (ACLs) of files.
CALL       Calls one batch program from another.
CD         Displays the name of or changes the current directory.
CHCP       Displays or sets the active code page number.
CHDIR      Displays the name of or changes the current directory.
CHKDSK     Checks a disk and displays a status report.
CHKNTFS    Displays or modifies the checking of disk at boot time.
CLS        Clears the screen.
CMD        Starts a new instance of the Windows command interpreter.
COLOR      Sets the default console foreground and background colors.
COMP       Compares the contents of two files or sets of files.
COMPACT    Displays or alters the compression of files on NTFS partitions.
CONVERT    Converts FAT volumes to NTFS. You cannot convert the
           current drive.
COPY       Copies one or more files to another location.
DATE       Displays or sets the date.
DEL        Deletes one or more files.
DIR        Displays a list of files and subdirectories in a directory.
DISKPART   Displays or configures Disk Partition properties.
DOSKEY     Edits command lines, recalls Windows commands, and
           creates macros.
DRIVERQUERY Displays current device driver status and properties.
ECHO       Displays messages, or turns command echoing on or off.
ENDLOCAL   Ends localization of environment changes in a batch file.
-- More --
```



When looking through the list, press the **Spacebar** to move down the list per page or the **Enter** key to go down the list per line. Press **Q** to quit viewing the list at any time.

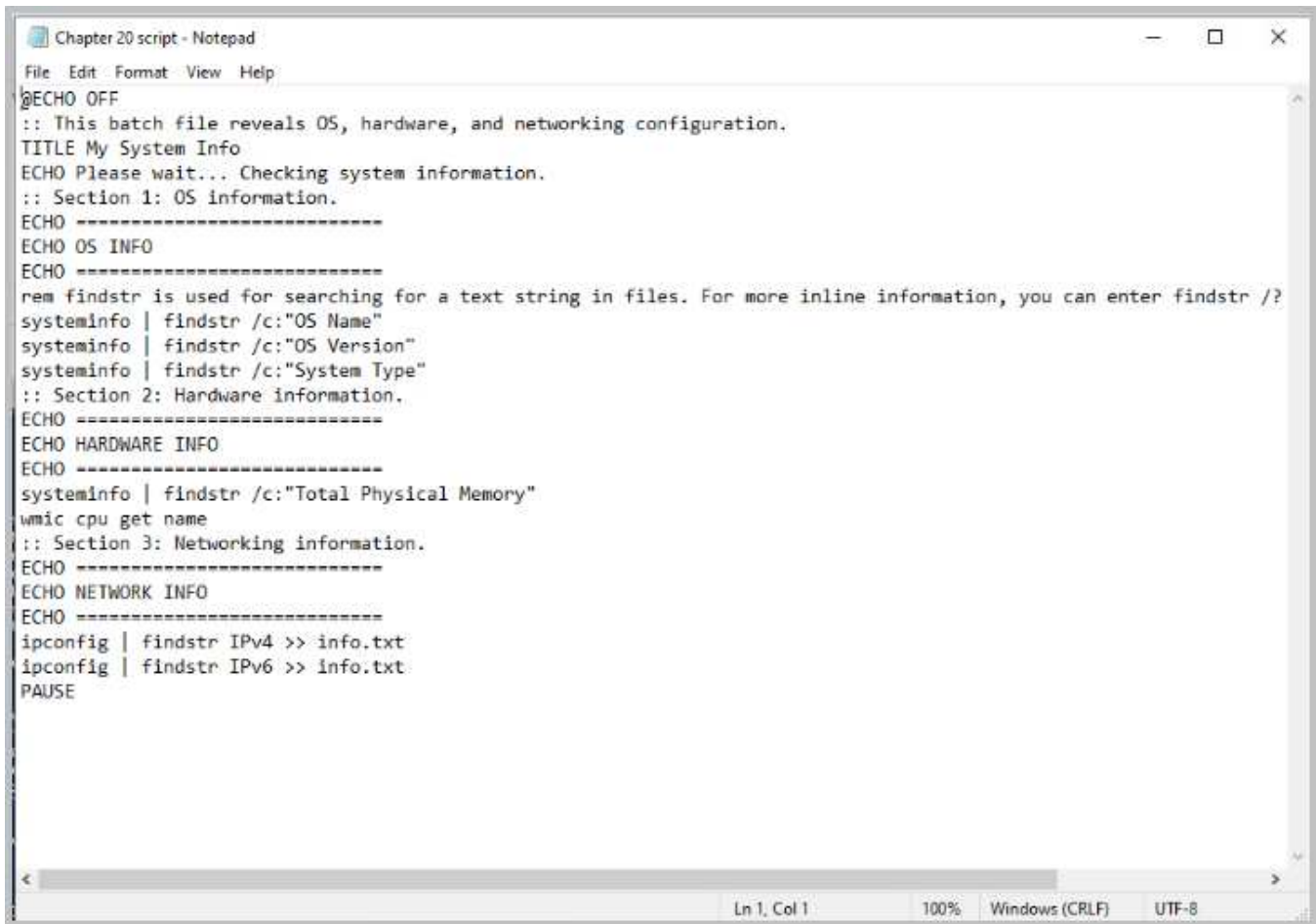
- Once you have finished reviewing the command prompt's output, in the **search** textbox, type **notepad** and click **Open**.





5. Enter the following text into the *Notepad* file.

```
@ECHO OFF
:: This batch file reveals OS, hardware, and networking configuration.
TITLE My System Info
ECHO Please wait... Checking system information.
:: Section 1: OS information.
ECHO =====
ECHO OS INFO
ECHO =====
rem findstr is used for searching for a text string in files. For more inline
information, you can enter findstr /? at the command prompt.
systeminfo | findstr /c:"OS Name"
systeminfo | findstr /c:"OS Version"
systeminfo | findstr /c:"System Type"
:: Section 2: Hardware information.
ECHO =====
ECHO HARDWARE INFO
ECHO =====
systeminfo | findstr /c:"Total Physical Memory"
wmic cpu get name
:: Section 3: Networking information.
ECHO =====
ECHO NETWORK INFO
ECHO =====
ipconfig | findstr IPv4 >> info.txt
ipconfig | findstr IPv6 >> info.txt
PAUSE
```



```
Chapter 20 script - Notepad
File Edit Format View Help
@ECHO OFF
:: This batch file reveals OS, hardware, and networking configuration.
TITLE My System Info
ECHO Please wait... Checking system information.
:: Section 1: OS information.
ECHO =====
ECHO OS INFO
ECHO =====
rem findstr is used for searching for a text string in files. For more inline information, you can enter findstr /?
systeminfo | findstr /c:"OS Name"
systeminfo | findstr /c:"OS Version"
systeminfo | findstr /c:"System Type"
:: Section 2: Hardware information.
ECHO =====
ECHO HARDWARE INFO
ECHO =====
systeminfo | findstr /c:"Total Physical Memory"
wmic cpu get name
:: Section 3: Networking information.
ECHO =====
ECHO NETWORK INFO
ECHO =====
ipconfig | findstr IPv4 >> info.txt
ipconfig | findstr IPv6 >> info.txt
PAUSE
```



### Breakdown of commands in sysinfo.bat

@ECHO – turns off ECHO

ECHO - print the command currently running on to the command prompt.

Rem- this does not execute it is a remark or note for the programmer

:: - is a comment

Findstr - is used for searching for a text string in files. For more inline information, enter findstr /? at the command prompt

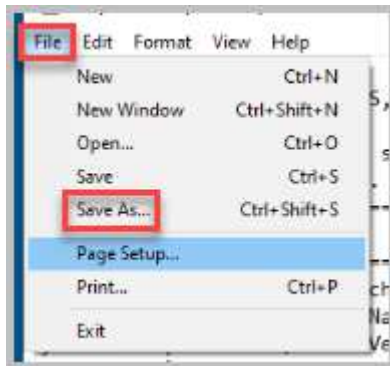
Ipconfig – display network information

>> redirects output to a file if no directory is indicated it is added to the current directory so you will not see the output in the command prompt it will have to be read from the file.

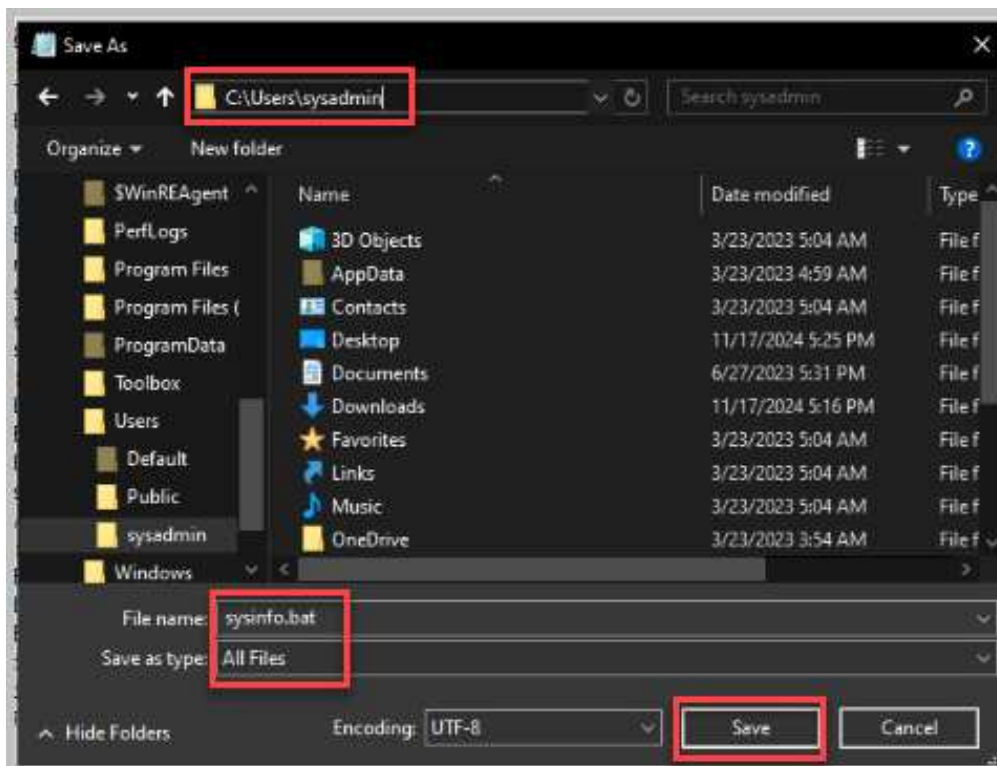
| (key located above the Enter key) - an operator that passes the output of the command before it in the line to the command on the right of it.

PAUSE – requires a key press to continue

6. Click **File > Save As**.

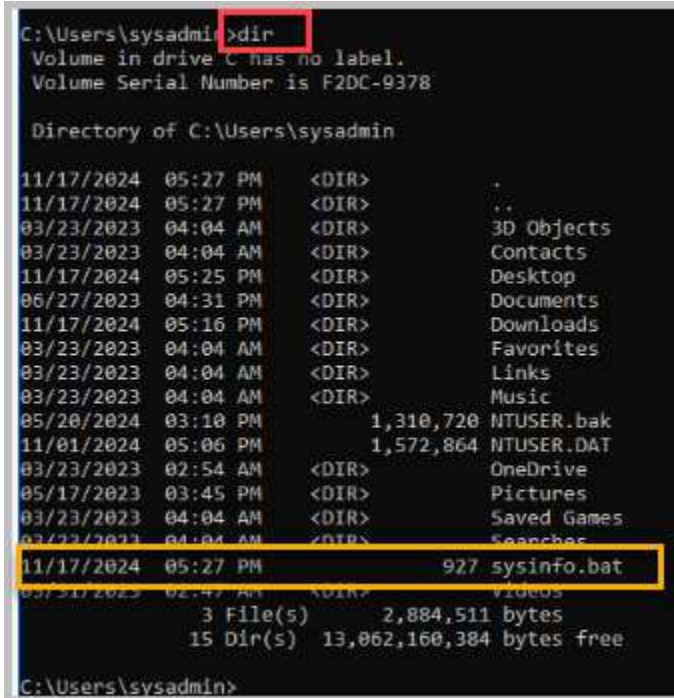


7. Name the file `sysinfo.bat` and save it in the `C:\Users\sysadmin` directory. Verify that the *Save as Type* is listed as **All Files**. Select **Save**.



8. Change focus back to the command prompt window and list the contents of the *sysadmin* directory by entering the command below to verify that the file *sysinfo.bat* is saved with the correct file extension. If not, rename the file to *sysinfo.bat*.

```
dir
```



```
C:\Users\sysadmin>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

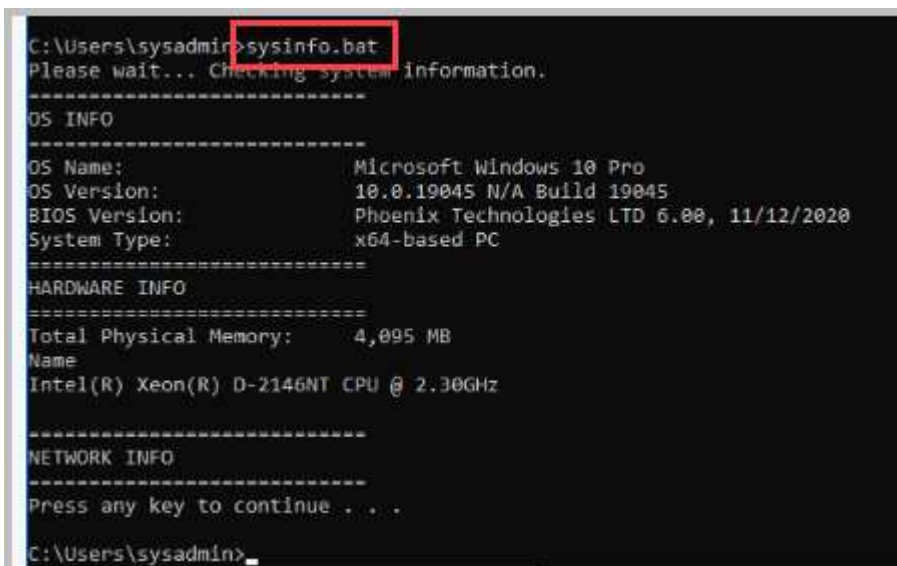
Directory of C:\Users\sysadmin

11/17/2024  05:27 PM    <DIR>          .
11/17/2024  05:27 PM    <DIR>          ..
03/23/2023  04:04 AM    <DIR>          3D Objects
03/23/2023  04:04 AM    <DIR>          Contacts
11/17/2024  05:25 PM    <DIR>          Desktop
06/27/2023  04:31 PM    <DIR>          Documents
11/17/2024  05:16 PM    <DIR>          Downloads
03/23/2023  04:04 AM    <DIR>          Favorites
03/23/2023  04:04 AM    <DIR>          Links
03/23/2023  04:04 AM    <DIR>          Music
05/20/2024  03:10 PM             1,310,720 NTUSER.bak
11/01/2024  05:06 PM             1,572,864 NTUSER.DAT
03/23/2023  02:54 AM    <DIR>          OneDrive
05/17/2023  03:45 PM    <DIR>          Pictures
03/23/2023  04:04 AM    <DIR>          Saved Games
03/23/2023  04:04 AM    <DIR>          Searches
11/17/2024  05:27 PM             927 sysinfo.bat
03/23/2023  02:47 AM    <DIR>          Videos
               3 File(s)      2,884,511 bytes
               15 Dir(s)  13,062,160,384 bytes free

C:\Users\sysadmin>
```

9. At the prompt, enter the command below to run the *sysinfo.bat* script. Carefully watch the output to follow the commands listed in the file. When prompted to press any key to continue, press any key to continue.

```
sysinfo.bat
```



```
C:\Users\sysadmin>sysinfo.bat
Please wait... Checking system information.
=====
OS INFO
=====
OS Name:             Microsoft Windows 10 Pro
OS Version:          10.0.19045 N/A Build 19045
BIOS Version:        Phoenix Technologies LTD 6.00, 11/12/2020
System Type:         x64-based PC
=====
HARDWARE INFO
=====
Total Physical Memory:  4,095 MB
Name
Intel(R) Xeon(R) D-2146NT CPU @ 2.30GHz
=====
NETWORK INFO
=====
Press any key to continue . . .
C:\Users\sysadmin>
```

10. List the content of the sysadmin directory to verify that the file *info.txt* has been created as a result of the script. Enter the command below.

```
dir
```

```
C:\Users\sysadmin>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin

11/17/2024  05:29 PM  <DIR>          .
11/17/2024  05:29 PM  <DIR>          ..
03/23/2023  04:04 AM  <DIR>          3D Objects
03/23/2023  04:04 AM  <DIR>          Contacts
11/17/2024  05:25 PM  <DIR>          Desktop
06/27/2023  04:31 PM  <DIR>          Documents
11/17/2024  05:16 PM  <DIR>          Downloads
03/23/2023  04:04 AM  <DIR>          Favorites
11/17/2024  05:29 PM             121 info.txt
03/23/2023  04:04 AM  <DIR>          Links
03/23/2023  04:04 AM  <DIR>          Music
05/20/2024  03:10 PM             1,310,720 NTUSER.bak
11/01/2024  05:06 PM             1,572,864 NTUSER.DAT
03/23/2023  02:54 AM  <DIR>          OneDrive
05/17/2023  03:45 PM  <DIR>          Pictures
03/23/2023  04:04 AM  <DIR>          Saved Games
03/23/2023  04:04 AM  <DIR>          Searches
11/17/2024  05:27 PM             927 sysinfo.bat
03/31/2023  02:47 AM  <DIR>          Videos
               4 File(s)              2,884,632 bytes
              15 Dir(s) 13,023,723,520 bytes free

C:\Users\sysadmin>
```

11. At the prompt, type the command **more info.txt** to see the file created by the script. Review the output of the command.

```
more info.txt
```

```
C:\Users\sysadmin>more info.txt
IPv4 Address. . . . . : 192.168.1.10
Link-local IPv6 Address . . . . . : fe80::719d:621e:e84b:df7%15

C:\Users\sysadmin>
```

12. Close the command prompt window.  
13. Close the Notepad window.  
14. Stay logged in to the *Win10-OS* machine to continue to the next section.

## 2 Creating and Testing a PowerShell Script

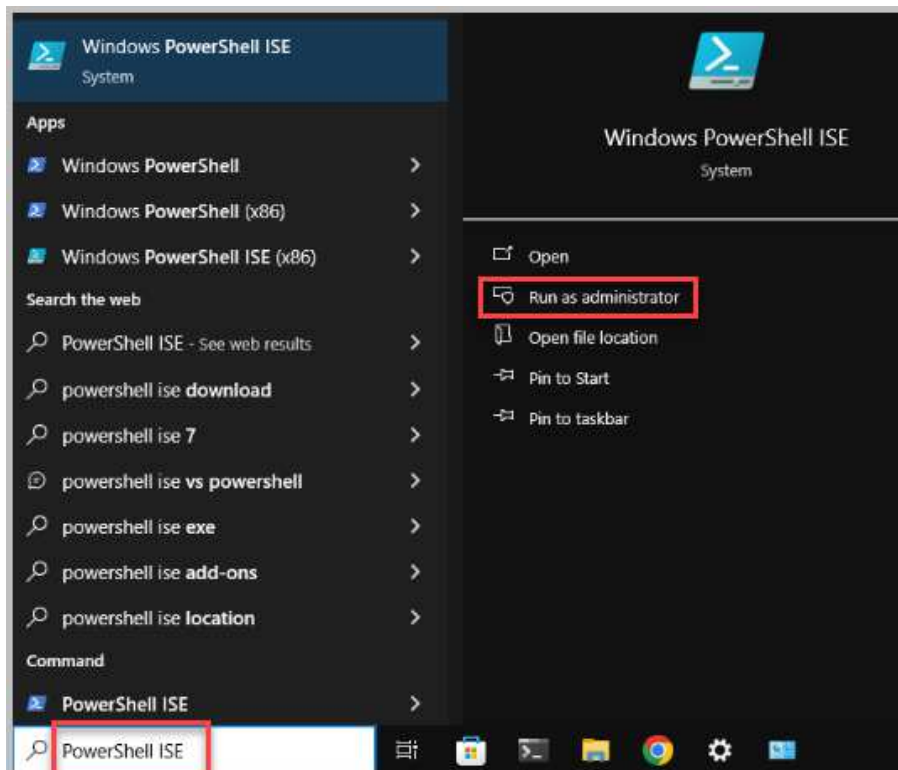
Windows PowerShell is a robust command-line shell and scripting language designed primarily for system administration on Windows. It empowers administrators to efficiently control and automate a wide range of tasks, from simple configuration changes to intricate system management operations.

PowerShell employs a unique approach, utilizing cmdlets (commandlets) instead of traditional commands. Cmdlets are powerful, versatile functions that provide a more structured and efficient way to interact with the system. By chaining cmdlets together in a pipeline, administrators can create complex automation workflows with minimal effort.

Batch files and PowerShell scripts are both collections of commands saved in text files. Batch files utilize the .bat extension and are typically used for simpler, sequential tasks. PowerShell scripts, on the other hand, employ the .ps1 extension and offer significantly greater flexibility and power due to their object-oriented nature and extensive scripting capabilities.

By default, Windows 10 has a security policy that prevents the automatic execution of PowerShell scripts. This safeguard is designed to protect systems from malicious scripts. To run a PowerShell script, administrators must adjust the execution policy to a more permissive setting. This can be done through the PowerShell console or by using the Run as Administrator option.

1. In the *Search Text Box* on the taskbar, type **PowerShell ISE**. Select **Run as administrator**.



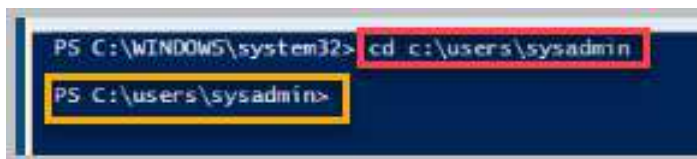


- Click **Yes** to accept the *User Account Control*.

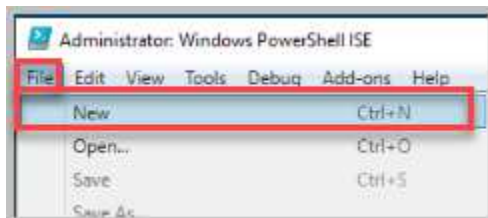


- At the *PowerShell ISE* prompt, navigate to the *sysadmin* home directory using the **cd** command. Type `cd c:\users\sysadmin` and select **Enter** on your keyboard.

```
cd c:\users\sysadmin
```



- In the *PowerShell ISE* window, click on **File**, then click **New** to create a new script.

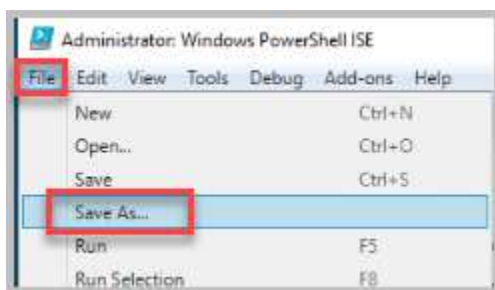


- Enter the following commands into the newly created *Untitled.ps1* window.

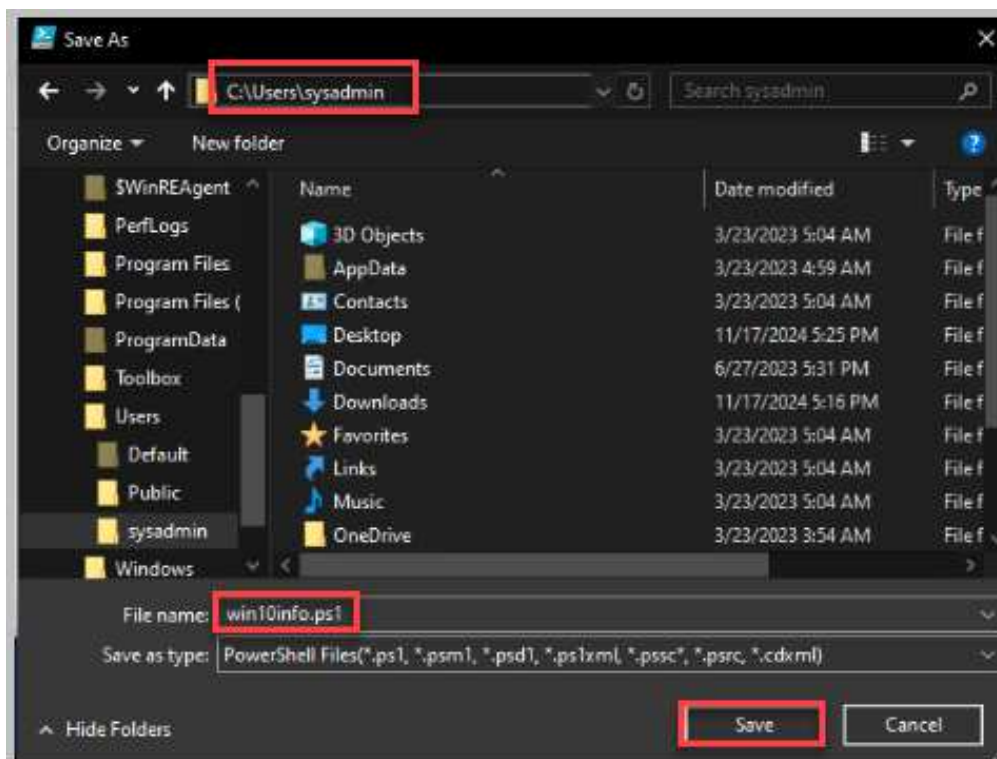
```
Write-Output "Computer name is:"
Get-Content env:computername
Write-Output "Windows version is:"
(Get-WmiObject -Class Win32_OperatingSystem).Caption
Write-Output "CPU is:"
Get-WmiObject Win32_Processor | findstr "Name"
Write-Output "Total Memory is:"
[Math]::Round((Get-WmiObject -Class Win32_ComputerSystem -ComputerName
localhost).TotalPhysicalMemory/1Gb)
Write-Output "The Disks that are installed and their freespace:"
Get-WmiObject -Class Win32_LogicalDisk -Filter "DriveType = '3'"
Write-Output "IPv4 addresses:"
Get-NetIPAddress -AddressFamily IPv4 | Sort-Object -Property InterfaceIndex |
Format-Table
```

```
Untitled1.ps1 X
1 Write-Output "Computer name is:"
2 Get-Content env:computername
3 Write-Output "Windows version is:"
4 (Get-WmiObject -Class Win32_OperatingSystem).Caption
5 Write-Output "CPU is:"
6 Get-WmiObject Win32_Processor | findstr "Name"
7 Write-Output "Total Memory is:"
8 [Math]::Round((Get-WmiObject -Class Win32_ComputerSystem -ComputerName localhost).TotalPhysicalMemory/1Gb)
9 Write-Output "The Disks that are installed and their freespace:"
10 Get-WmiObject -Class Win32_LogicalDisk -Filter "DriveType = '3'"
11 Write-Output "IPv4 addresses:"
12 Get-NetIPAddress -AddressFamily IPv4 | Sort-Object -Property InterfaceIndex | Format-Table
```

6. Once all the lines are typed, click **File** on the menu bar and select **Save As**.



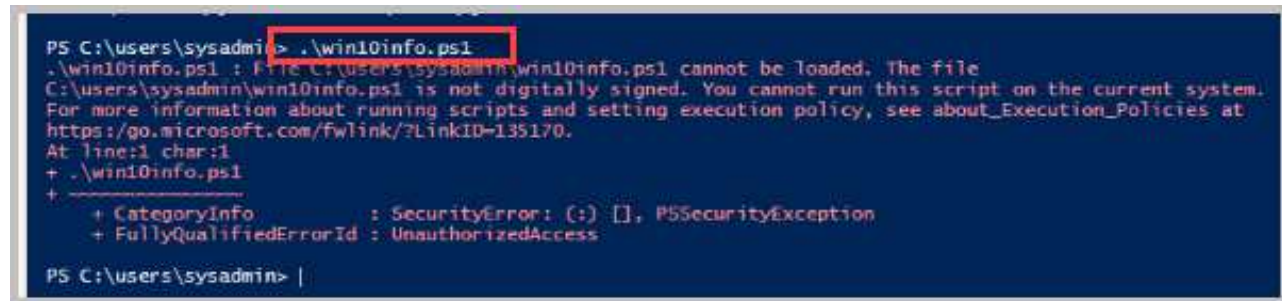
7. In the **Save As** window, name the file **win10info.ps1**, and be sure to save it to the **C:\Users\sysadmin** home directory.





8. Back on the *PowerShell ISE* prompt, type `.\win10info.ps1` command, followed by pressing the **Enter** key to run the script.

```
.\win10info.ps1
```



```
PS C:\users\sysadmin> .\win10info.ps1
.\win10info.ps1 : File C:\users\sysadmin\win10info.ps1 cannot be loaded. The file
C:\users\sysadmin\win10info.ps1 is not digitally signed. You cannot run this script on the current system.
For more information about running scripts and setting execution policy, see about_Execution_Policies at
https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ .\win10info.ps1
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess

PS C:\users\sysadmin> |
```

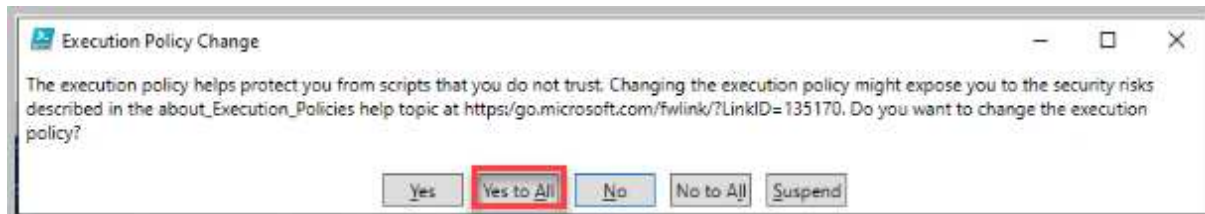
9. Notice that an error appears signaling that this script is not digitally signed and that scripts are disabled on the system by default. To disable the security feature, enter the command below.

```
Set-ExecutionPolicy RemoteSigned
```



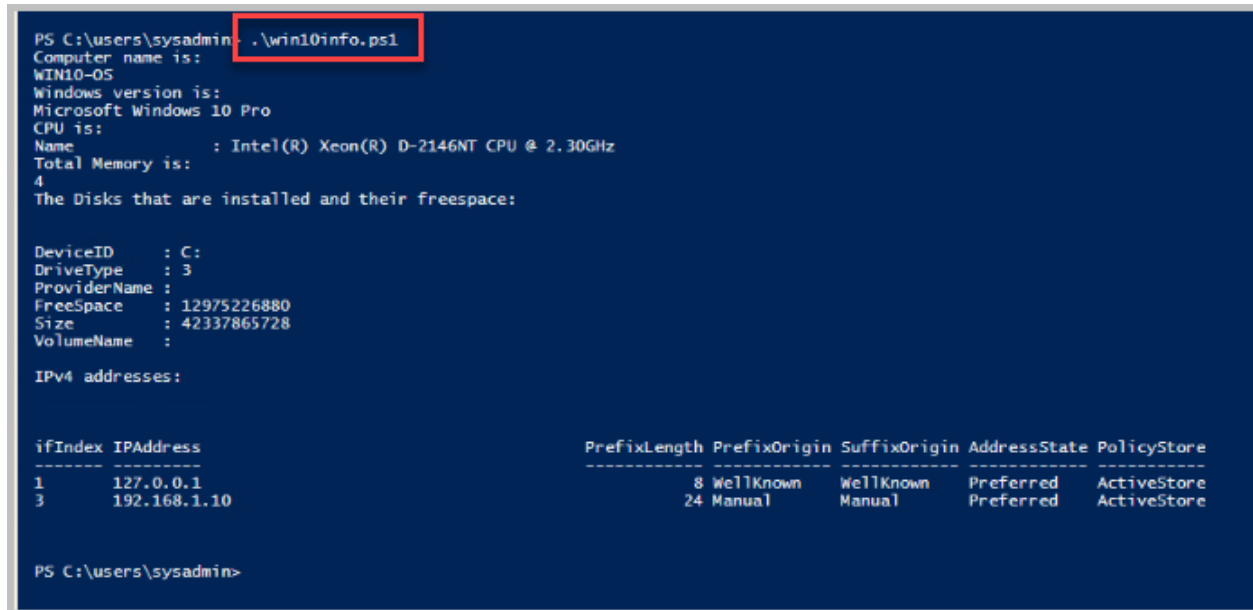
```
PS C:\users\sysadmin> Set-ExecutionPolicy RemoteSigned
```

10. A pop-up window appears. Click **Yes to All** to continue.



11. Back on the *PowerShell ISE* prompt, reenter the command below to launch the script.

```
.\win10info.ps1
```



```
PS C:\users\sysadmin> .\win10info.ps1
Computer name is: WIN10-05
Windows version is: Microsoft Windows 10 Pro
CPU is:
Name       : Intel(R) Xeon(R) D-2146NT CPU @ 2.30GHz
Total Memory is: 4
The Disks that are installed and their freespace:

DeviceID   : C:
DriveType  : 3
ProviderName :
FreeSpace   : 12975226880
Size       : 42337865728
VolumeName :

IPv4 addresses:

ifIndex  IPAddress      PrefixLength PrefixOrigin SuffixOrigin AddressState PolicyStore
-----
1        127.0.0.1      8 WellKnown   WellKnown   Preferred ActiveStore
3        192.168.1.10  24 Manual     Manual     Preferred ActiveStore

PS C:\users\sysadmin>
```

12. Review the output of the script. What comparisons do you see between the Batch file and the PowerShell script? Notice that *ECHO* in the batch file is similar to *Write-Output* in *PowerShell (PS)*. Also, *%computername%* in the batch represents a variable like *Get-Content* reads the *env:computername*. CPU is: *%PROCESSOR\_IDENTIFIER%* is like *Get-WmiObject Win32\_Processor | findstr "Name"*. Try to find the other batch commands that are equal the PowerShell commands and cmdlets.

### 3 Creating a BASH Script using Linux Mint and Testing a Bash Script

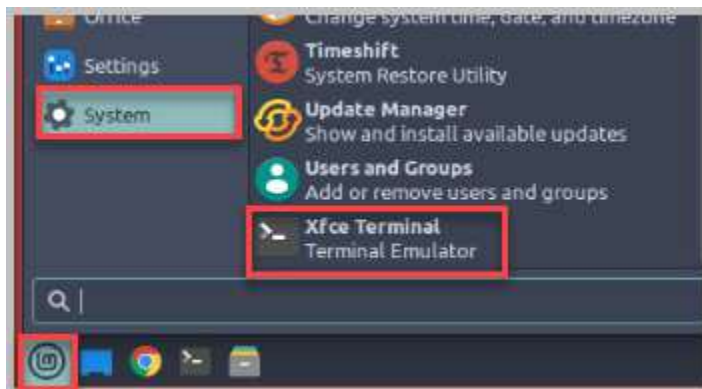
*BASH* is a command interpreter used in Linux and Unix distributions (distros). *Windows Subsystem for Linux* is now also available as a feature in *Windows 10 x64* only. According to Microsoft, “*Windows Subsystem for Linux* enables you to run native *Linux* command-line tools directly on *Windows*, alongside your traditional *Windows* desktop and modern store apps,” and it allows you to use the *BASH* shell. *BASH* shell is the environment you type commands that will enable you to interact with your computer. It is a way for you to have the computer do a task, calculation, or something you want it to do. *BASH* shell scripting is a way to use the *BASH* shell interaction to automate command execution with a single command or group of commands.

In this exercise, you will create a *BASH* script and use the *BASH* shell on the *Linux-OS* system to run it. To create a *BASH* script like a batch file in *Windows*, you will need to use a text editor. In *Windows 10*, you use *Notepad*; on the *Linux-OS* system, you will use a built-in text editor tool called *nano*.

1. Access the **Linux-OS** system from the pod topology.

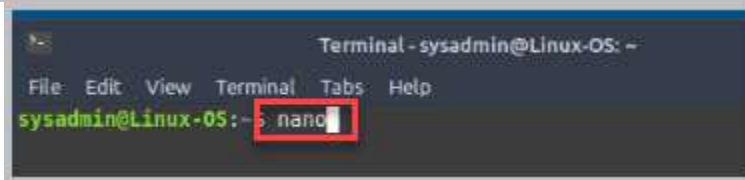


2. Click on the **Applications** button located on the bottom taskbar and navigate to **System > Xfce Terminal** to open the terminal window, which, in comparison to *Windows*, is similar to the command prompt with regards to using the command-line interface.

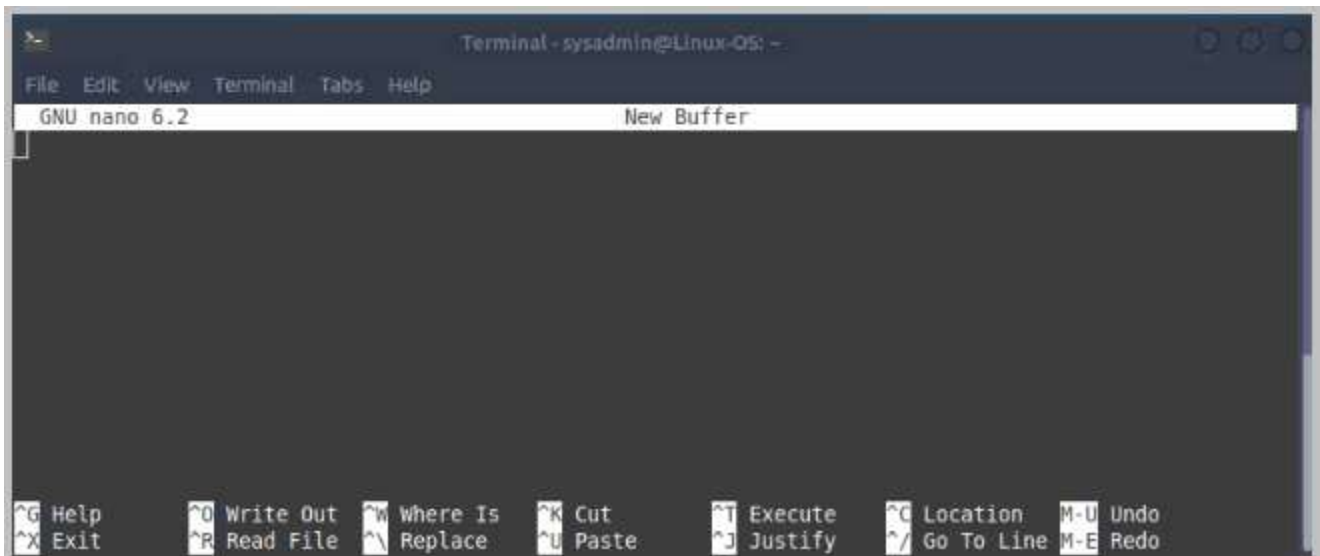


- At the prompt, type the command below, followed by pressing the **Enter** key to open the nano text editor.

nano

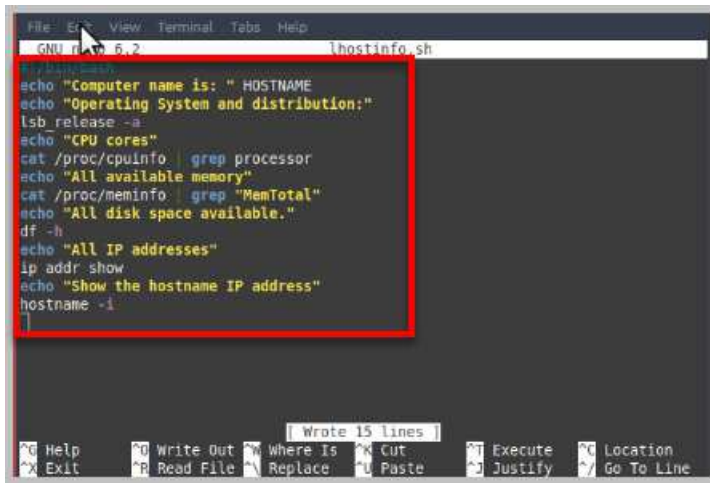


- GNU nano 6.2 is a user-friendly text editor designed for the command-line environment. It's particularly popular on Unix-like systems, including Linux and macOS.



- You will use the *nano* text editor tool to create a file named *lhostinfo.sh* with the following text. Then you will save it in your home directory. First, start by creating the file by typing the following lines in the *nano* text editor.

```
#!/bin/bash
echo "Computer name is: " HOSTNAME
echo "Operating System and distribution:"
lsb_release -a
echo "CPU cores"
cat /proc/cpuinfo | grep processor
echo "All available memory"
cat /proc/meminfo | grep "MemTotal"
echo "All disk space available."
df -h
echo "All IP addresses"
ip addr show
echo "Show the hostname IP address."
hostname -i
```



```

File Edit View Terminal Tabs Help
GNU nano 6.2 lhostinfo.sh
#!/bin/bash
echo "Computer name is: " HOSTNAME
echo "Operating System and distribution:"
lsb_release -a
echo "CPU cores"
cat /proc/cpuinfo | grep processor
echo "All available memory"
cat /proc/meminfo | grep "MemTotal"
echo "All disk space available."
df -h
echo "All IP addresses"
ip addr show
echo "Show the hostname IP address"
hostname -i

```



**#!/bin/bash** - **#!** Called shebang, it is the syntax used in scripts to indicate an interpreter for execution in this case bash.

**lsb** – a utility that displays Linux Standard Base information about the Linux distribution.

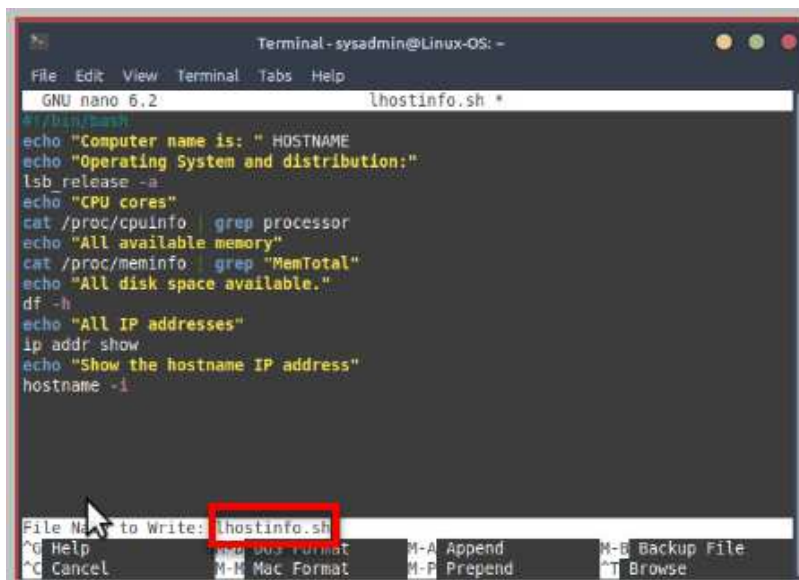
**cat** – short for concatenate, is a command which reads data from the file and gives the content as output.

**/proc** – virtual files that contain dynamic information about the system.

**df** – short for disk filesystem, this command will get a summary of the available and used disk space on your system.

**ip addr** – Change to show all IP addresses on the system.

- To save a file, use the keyboard shortcut **Ctrl+O**. With this key combination, the editor will ask you to provide a filename. Use **lhostinfo.sh** to name your file.



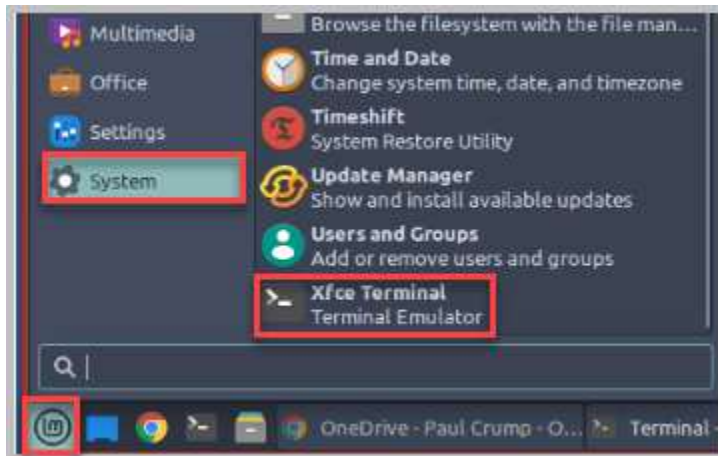
```

Terminal - sysadmin@Linux-OS: -
File Edit View Terminal Tabs Help
GNU nano 6.2 lhostinfo.sh *
#!/bin/bash
echo "Computer name is: " HOSTNAME
echo "Operating System and distribution:"
lsb_release -a
echo "CPU cores"
cat /proc/cpuinfo | grep processor
echo "All available memory"
cat /proc/meminfo | grep "MemTotal"
echo "All disk space available."
df -h
echo "All IP addresses"
ip addr show
echo "Show the hostname IP address"
hostname -i
File Name to Write: lhostinfo.sh

```

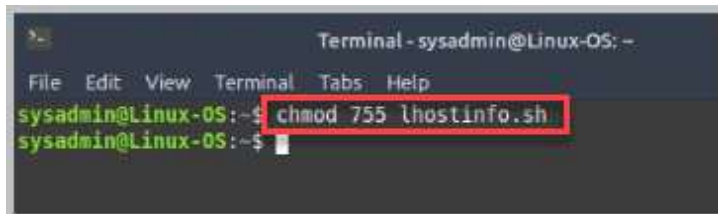
- Press **Enter** to save the file to the sysadmin's home directory.

8. Open a second terminal by selecting the **Start** button, **System > Xfce Terminal**.



9. You will need to make the script executable. Enter the command below in the newly opened terminal. Press **Enter** to save the file permission changes.

```
chmod 755 lhostinfo.sh
```



10. In the same terminal, enter the command below to execute the script. You will need to use `./` to tell the `BASH` interpreter where the script is located. There is a fixed order in which the `BASH` interpreter checks directories to find and run a command. If your script is not in one of these directories, then it will not find your script unless you specify the path. The `./` is telling the `BASH` interpreter that the script is in the current directory, but you could also run the script from any other directory by using the full path. Press **Enter** to execute the command.

```
./lhostinfo.sh
```

```
sysadmin@Linux-05:~$ ./lhostinfo.sh
Computer name is: h0stn0v4
Operating System and distribution:
No LSB modules are available.
Distributor ID: Linuxmint
Description:   Linux Mint 21
Release:      21
Codename:     vanessa
CPU cores
processor      : 8
All available memory
MemTotal:     2011084 kB
All disk space available.
Filesystem    Size  Used Avail Use% Mounted on
tmpfs         197M  10M  187M   6% /run
/dev/mapper/vgmint-root 38G   11G   26G  29% /
tmpfs         902M   0  902M   0% /dev/shm
tmpfs         5.0M   0   5.0M   0% /run/lock
/dev/sda2     512M  6.1M  506M   2% /boot/efi
tmpfs         197M  112K  197M   1% /run/user/1000
All IP addresses
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid lft forever preferred_lft forever
2: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:56:b3:35:e4 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
    inet 192.168.1.7/24 brd 192.168.1.255 scope global noprefixroute ens192
        valid lft forever preferred_lft forever
    inet6 fe80::90c6:e86f:2449:9908/64 scope link noprefixroute
        valid lft forever preferred_lft forever
Show the hostname IP address
127.0.1.1
sysadmin@Linux-05:~$
```

11. The lab is now complete; you may end the reservation.